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The value of data, machine learning, and deep learning in restaurant demand forecasting: Insights and lessons learned from a large restaurant chain

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ABSTRACT

The restaurant industry has been slow to adopt analytics for the supply chain, operations, and demand forecasting, with limited research on this sector. The COVID-19 pandemic's significant impact on the restaurant industry, one of the hardest-hit sectors, has underscored the need for digital technologies and advanced analytics for managing supply chains and making operational decisions. This paper presents a collaborative study with one of the largest restaurant chains in the United States, highlighting the value of advanced data analytics in forecasting restaurant demand. The study offers insights into the benefit of integrating external data, including macroeconomic and pandemic-related factors, into demand forecasting. It explores traditional machine learning algorithms and state-of-the-art deep learning architectures, evaluating their effectiveness in the context of the restaurant industry. The paper further discusses the implications of utilizing advanced forecasting models, providing valuable insights for the restaurant industry in the face of supply chain disruptions and pandemics.

1. Introduction

Restaurants today face various challenges, such as increased supply chain uncertainty and high stakeholder expectations. The uncertainty of the restaurant business is compounded by unforeseen crises, such as the COVID-19 pandemic, along with fluctuating economic conditions, underscores the critical need for agility and adaptability in demand forecasting practices [37]. These challenges have prompted restaurants to accelerate the digitalization of customer demand forecasting, employing analytics with various information sources, both internal and external [15]. Consequently, there has been a growing awareness of the necessity for digital technologies and advanced analytics in supply chain and operational decision-making.

The Dynamic Capabilities Framework provides an effective approach to address the challenges faced by the restaurant industry. This framework emphasizes the dynamic capabilities of organizations to identify, develop, and modify internal and external competencies, enabling them to navigate rapidly changing environments [18]. The perspective of this study revolves around three dynamic capabilities: *sensing* opportunities and risks, *seizing* capabilities, and *transforming* operations to adapt to environmental changes [55]. This framework highlights the importance of advanced analytics in improving forecasting accuracy and developing strategic foresight by exploring both internal and external factors affecting a business. This study focuses on the need for advanced demand forecasting methods in the restaurant sector to navigate the intricate interplay of internal and external factors affecting demand.

Based on the Dynamic Capabilities Framework, this paper examines the development of demand forecasting through advanced analytic techniques in collaboration with one of the largest restaurant chains in the United States. The restaurant chain's supply chain operations are complex and characterized by a system in which heterogeneous elements interact, and the outcomes of such interactions are difficult to predict [50]. The pandemic has exacerbated the difficulty of managing such systems, and data and analytical techniques have emerged as a critical strategy to mitigate these challenges [15,28,45,52]. An executive in charge of the chain's supply chain operation and data team underscored the heightened uncertainty during these turbulent times as follows:

> The worldwide pandemic has disrupted the food supply chain in ways that were previously unimaginable. A system that once relied on predictability and "just-in-time" processing is facing a new reality

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where continuity of supply is anything but guaranteed. Wild demand swings and constrained production have led to national product shortages. For these reasons, restaurant supply chain organizations are rethinking inventory management. Tools and methods like machine learning, data visualization, and integrated business planning are being explored to create a real-time picture of promotions and product movement shared across all stakeholders.

To address the above issue, we analyzed a significant period (January 2019 – June 2022) of internal competencies of the restaurant firm (e.g., menu-level sales and promotion data from this chain) alongside external data such as macroeconomic indicators and pandemic-related variables. Both internal and external data were curated and processed for use in developing demand forecasting models using both ML and DL algorithms to test the roles of internal and external competencies in both stable and turbulent periods. The sales data illustrate how an unprecedented risk, such as the COVID-19 pandemic, affected the sales of a major U.S. restaurant chain and how the firm rebounded over time, showcasing the dynamics of the restaurant industry. In collaboration with the chain, this study addresses three research questions (RQs) to understand the value of data and advanced analytics for restaurant demand forecasting:

RQ1: How can restaurants enhance their sensing capabilities by integrating external data, such as macroeconomic indicators and pandemic-related variables?

RQ2: How do advanced analytics, specifically machine learning (ML) and deep learning (DL) techniques, contribute to strengthening restaurants' seizing and transforming capabilities in the context of restaurant demand forecasting?

RQ3: How do ML and DL algorithms differ in their impact on improving restaurants' ability to forecast demand and adapt to market changes?

RQ1 delves into how incorporating these non-traditional, external data sources, such as macroeconomic indicators and pandemic-related variables, can enhance the sensing capability of restaurants [62]. Traditionally, restaurants have primarily relied on internal data, particularly past sales data, for demand forecasting. The pandemic served as a wake-up call for the chain to broaden the types of data for more accurate demand forecasting, which leads to exploring the value of external data in the context of restaurants. RQ2 builds on the potential opportunities of utilizing advanced analytics, specifically ML and DL, in the restaurant context [25,28]. There has been considerable academic interest in applying AI techniques and methods, including ML and DL, within operations and supply chains [5]. RQ2 aims to understand the implications of adopting such advanced analytics to seize opportunities and transform operations in response to demand forecasting insights. Finally, RQ3 centers on the comparative value of DL-based demand forecasting models over conventional ML algorithms. With DL considered the state of the art in the AI domain and ML being well-established as a widely adopted technology, RQ3 seeks to compare their relative effectiveness in navigating market dynamics and fostering strategic foresight [33].

The remainder of this manuscript is outlined as follows: This study includes a review of academic literature on demand forecasting, primarily in restaurant contexts, a survey of data sources and ML and DL algorithms, and a comparison and discussion of model results. The discussion addresses the aforementioned research questions based on the outcomes of various models using different combinations of ML/DL algorithms and internal and external data in two business environments: stable and turbulent. Our findings are expected to contribute to the restaurant analytics literature and, more importantly, the restaurant business community by providing insights into deploying ML/DL-based demand forecasting.

2. Relevant literature

This section covers the restaurant industry's background, demand forecasting practices, data sources, and ML algorithms discussed in relevant literature.

2.1. Demand forecasting in the restaurant context

Food sold in restaurants is unique because it is prepared upon order and cannot be stored for long [10]. In addition, customer visits exhibit patterns based on certain factors, such as the day of the week, events, macroeconomic conditions, and promotional menus. Predicting customer demand accurately can reduce food materials and labor costs, leading to higher profits [35]. Inaccurate forecasting can result in wasted resources and business failure [51]. Therefore, demand forecasting is crucial for meeting customer needs, reducing costs, and maintaining adequate inventory.

Restaurants act as a bridge between producers and customers, involving both manufacturing and service supply chains [52]. Therefore, supply chain challenges can arise from the manufacturing side (e.g., supply disruption) [47] and the demand side (e.g., demand swing) [14]. In this dynamic environment, restaurant operations must balance resource management (upstream) and demand planning (downstream). Information from the downstream supply chain is crucial for upstream supply chain planning. Demand forecasting is a critical component of such resource planning, which includes inventory replenishment and distribution.

The restaurant industry has lagged behind other sectors (e.g., consumer goods and manufacturing) in adopting technologies and analytics for operation and decision support [23]. The industry has relied on qualitative data, such as managers' intuition, informed by various social orientation variables and unsophisticated demand forecasting methods. However, digital innovation has increased the use of data and advanced analytics, especially among large restaurant chains [52]. Nonetheless, limited research on restaurant analytics, particularly multivariate restaurant demand forecasting, indicates a critical research gap [43].

2.2. Relevant data for restaurant demand forecasting

Demand forecasting accuracy depends on data availability and granularity [1,5]. With the ongoing digital transformation in the restaurant industry and society, the demand forecasting data is becoming increasingly diverse and voluminous. This data can be categorized into data sources and time-related features (Fig. 1). Data sources span a spectrum of internal and external channels, each contributing unique insights. Technological advancements, exemplified by Point of Sales (POS) systems, have expanded the range of available data sources. Internal data encompasses crucial information, such as menu prices and marketing efforts, offering insights directly tied to business operations. External data comprises statistical and public data, reflecting broader social characteristics. This includes demographic information and weather conditions, highlighting the contextual factors influencing consumer behavior.

Time-related features introduce a layer of categorization. Some data exhibit stability over time, such as menu categories and county-level incomes. These elements, considered constants, provide a foundational backdrop for analysis. However, dynamic variables, such as consumer confidence levels and evolving health regulations and policies, demand a more adaptive approach.

Table 1 summarizes previous studies on restaurant demand forecasting, focusing on the data considered in the studies. Previous studies vary depending on different aspects, such as the demand forecasting unit, store or corporate level, internal and external data types employed [15,62], and the duration of data samples used for model training and validation.

Previous studies have focused on store or merchant-level demand

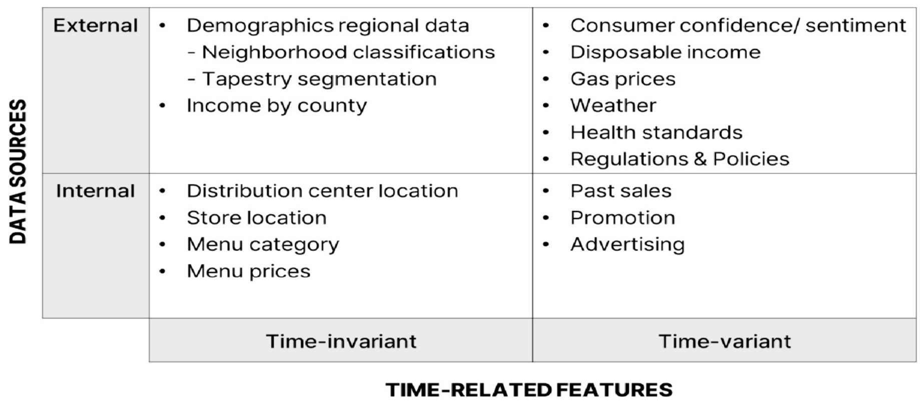

Fig. 1. Types of Data in Restaurant Demand Forecasting.

Table 1
Summary of restaurant demand forecasting literature.

Source	Study focus	Internal input data	External input data	Forecasting method	Model validation
Fernandes, Moro, Cortez, Batista and Ribeiro [15]	- Sample: Six restaurants - Data duration: N/A - Output: Monthly customer visits per restaurant	- Historical sales	- Online reviews (Tripadvisor)	- Statistical method (e.g., Bass model)	- N/A
Meneghini, Anzanello, Kahmann and Tortorella [34]	- Sample: Two types of hamburger - Data duration: 75 days of hamburger consumption - Output: Daily consumption of two types of hamburger	- Historical demand	- Expert opinions	- Statistical method (e.g., Moving average)	- Split validation (60 days for training and 15 days for testing)
Pandey, Mishra, Kumar, Kumar, Diwakar and Tripathi [36]	- Sample: A public dataset - Data duration: 145 weeks - Output: Weekly demand for food items	- Historical demand - Price - Location - Promotion	- N/A	- ML (Random forest regressor, Gradient boosting)	- Split validation (139 weeks for training and six weeks for testing
Schmidt, Kabir and Hoque [46]	- Sample: A restaurant - Data duration: Three years of restaurant gross sales - Output: One-day gross sales & one-week gross sales	- Historical sales - Date and time	- N/A	- ML - DL (Recurrent Neural Network)	- Window-base split validation (last one-day and one-week sales as testing)
Tanizaki, Hoshino, Shimmura and Takenaka [53]	- Sample: Visitor data for five stores of a restaurant chain - Data duration: Two years of visitor data - Output: A year visitor for each store	- Historical visitor data - Months and days - Holidays	- Weather (e.g., Temperature)	- ML	- Split validation (one year for training and one year for testing)
Tanizaki et al. [54]	- Sample: Beer demand of four stores from a restaurant chain - Data duration: Three years - Output: A year beer order	- Historical visitor data - Months and days - Holidays	- Weather (e.g., Temperature)	- ML	- Split validation (two years for training and one year for testing)
Yang, Liu and Chen [62]	- Sample: US country-level restaurant visit - Data duration: Three months - Output: US county-level restaurant visits	- N/A	- Weather - Covid-19 cases and policy - County-level data (e.g., population)	- Econometric model	- N/A

forecasting. Tanizaki et al. [54] employed POS data from four restaurant stores, supplemented with secondary data on weather and events. They aimed to forecast customer order quantity and inventory requirements for specific menu items at the individual store level. In such analyses, supply planning is assumed to be local. Local factors, such as events, weather conditions, online customer reviews [15], and location demographics, are considered. This is called the "centralized approach," where a single model is developed to forecast an aggregated total of all stock-keeping units. While this approach is straightforward and helpful for sales planning, a large restaurant chain requires a decentralized approach for forecasting individual menu items. This decentralized approach can be more complex and computationally intensive. Nevertheless, menu-level forecasting is essential for effective supply chain management, encompassing procurement, fulfillment, distribution, and inventory management [26].

In addition, previous studies relied on a small set of internal and external data for model training and validation. Pandey et al. [5] utilized only internal data, such as historical demand and promotion information. Schmidt et al. [6] used a restaurant's historical sales and time-related variables for building models, which were evaluated by comparing the last one-day and one-week sales forecasts. Three studies [2,7,8] in Table 1 employed internal and external data; however, the diversity of external features was limited to online ratings and weather. The pandemic-related features were considered in only one study [9], which focused on understanding county-level restaurant visits in the United States rather than forecasting restaurant demand at the store or corporate level.

Furthermore, studies evaluated model performance over short time frames (e.g., a few days) [9]. However, for a forecasting model to be practical, its performance must be reliable for long-term predictions spanning multiple prediction periods (e.g., several weeks). This multi-horizon forecasting provides the information necessary for procurement, inventory management, and distribution planning. In response, our study incorporates several macroeconomic indicators and pandemic-related variables and investigates how these external data enhance the sensing capability of a large restaurant chain.

2.3. ML and DL algorithms

There is an extensive body of literature on forecasting theory that covers several methods and techniques. These vary in complexity and reference field, ranging from judgment, statistical, and econometric models to Bayesian forecasting and AI models [41]. Recently, AI models, including ML and DL, have gained significant attention. ML has been widely discussed and applied in decision support and operations [1,9,11,28,57,64,65]. ML formulates demand forecasting as a regression problem. Three types of ML algorithms deserve particular attention – regularized regressions, decision trees (DT), and support vector regressors (SVR). These ML algorithms were designed for conventional regression problems without the temporal order of observations in a time series data. In addition, DL has received much attention recently, as neural network-based models can uncover complex, nonlinear patterns from various data, including texts, images, and videos. Consequently, DL applications in forecasting have increasingly appeared in the relevant literature. Three types of DL algorithms are gaining popularity in time series analysis – multilayer perceptrons (MLP), recurrent neural networks (RNN), and transformers. While these DL algorithms differ in architecture and other specifics, they preserve long-term dependencies in time series data: Table 2 lists ML and DL algorithms and their relevant literature.

Only a few studies on restaurant demand forecasting use ML and DL algorithms [46]. As summarized in Table 1, researchers have relied on training a small number of statistical or ML-based models using relatively short-term data from a restaurant or multiple stores. The performance of these models is validated using a one-time split validation method. However, this approach assumes that the performance of the analytical models is stable and does not consider business contexts. In our study, we develop 20 ML- and DL-based models and evaluate their effectiveness for two business contexts – stable and turbulent.

Table 2
ML and DL Algorithms and Relevant Literature.

	Types	Algorithms	Key characteristics	Relevant literature
ML	Regularized regressions	Lasso Ridge	Simultaneously select important variables and fit multivariate data	[8,56]
	Decision trees	Random forest (RF) Gradient Boosting Machine (GBM), Light GBM, XGBoost	Utilizing ensemble techniques, combining multiple trees to make a more robust regression model	[5, 11, 19, 27, 67]
	Support vector machine	Support vector regressor (SVR)	Proven particularly useful in uncovering nonlinear relationships	[13,63]
DL	Multilayer Perceptrons (MLP)	Neural Hierarchical Interpolation for Time Series (NHITS)	Utilizing a forked MLP architecture that takes a sequence of time series data as input and generates output is considered the first “pure” DL algorithm that outperformed traditional statistical approaches in time series competitions such as M3 and M4	[2,8,41]
	Recurrent Neural Networks (RNN)	DeepAR (Probabilistic Forecasting with Autoregressive Recurrent Networks)	Based on the concept of autoregression, where past values influence future values and are designed by Amazon	[2,51]
	Transformers	Temporal Fusion Transformer (TFT)	Incorporating features such as gating mechanisms, temporal processing, variable selection networks, multi-horizon forecasting, and attention mechanisms	[36,62]

3. Data and methods

3.1. Data

3.1.1. Operational data

Restaurant chains regularly introduce new menus, referred to as “promotional menu plates.” Each promotional period lasts several weeks, and multiple new menu plates are introduced during each period. Thus, the restaurant menu plates follow evolutionary patterns discussed in the new product/service development literature [48]. New menu plates are introduced and evaluated based on customer demand, making some short-lived while others are more enduring.

The data we received from the industry partner encompassed nearly four years (January 2019–July 2022) of sales, promotional data, and internal forecasting. Specifically, the sales data contains the daily sales amounts of over several hundred menu plates. The promotional data indicates which menu plates are on promotion, with roughly four to six promotional periods each year, during which several menu plates are introduced or reintroduced. The industry partner provided our research team with pre-pandemic sales data to understand the impact of the pandemic on their business.

We learned how this restaurant chain has forecasted customer demand over the years. The forecasting approach is relatively simple. When a new menu plate is introduced, such as a steak-hamburger combo, the sales of a similar, previously introduced menu plate are used as the forecast amount for the new menu plate. As shown in Fig. 2, such forecasts are made at the beginning of the promotion, and the forecast amount remains unchanged for the duration of the menu item’s life.

3.1.2. Time-related features

Customer demands vary by time, making time-related variables essential for accurate forecasting [24]. Restaurant sales fluctuate, typically increasing during weekends and holidays. Previous studies [31] show that sales may surge during certain days, weeks, or months, often called “seasonality.” Fig. 3 illustrates the impact of holidays on the restaurant chain’s sales, displaying two years of sales data (2019 and 2021), with vertical lines representing nine United States-based holidays, such as Thanksgiving and Christmas. The figure demonstrates the positive impact holidays have on sales.

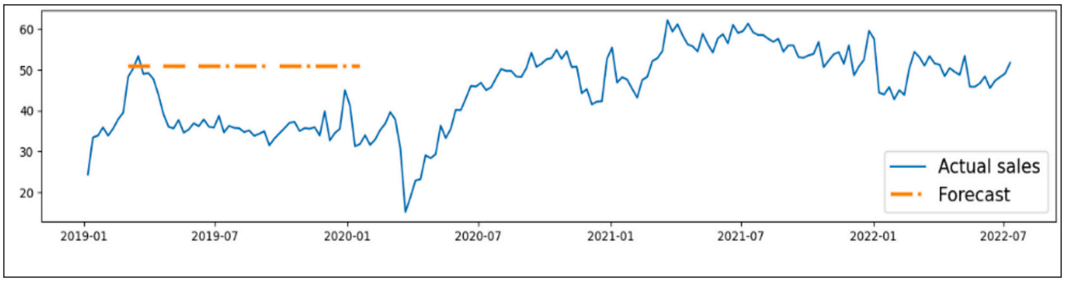

Fig. 2. The Chain's Forecasting Practice for Menu Sales.

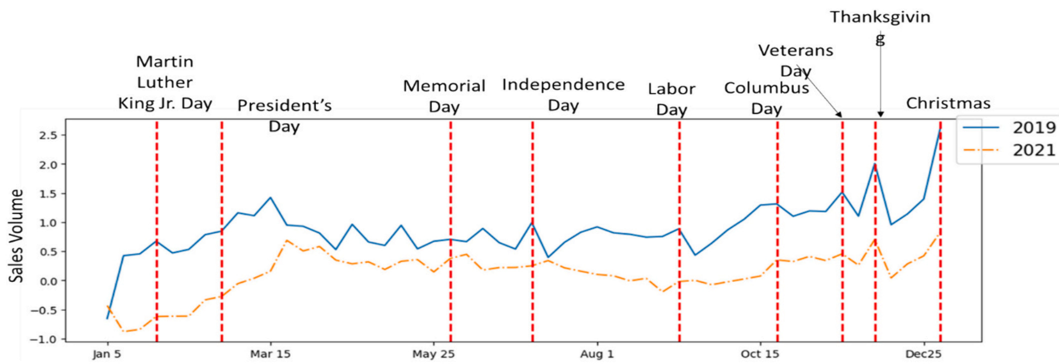

Fig. 3. Restaurant Sales During Regular Periods and United States Holidays.
Note. The dotted vertical lines represent holidays in the United States, and the vertical axis represents the normalized sales volume.

3.1.3. Macroeconomic and health policy features (MH)

Based on the advice of the restaurant's data team, we included two additional features – gas prices and people's moods or sentiments. To our knowledge, no study has empirically examined this relationship. However, a previous study investigating the relationship between macroeconomic indicators and the United States restaurant industry reported a negative impact of the consumer price index (CPI) on the sector [27]. Our partner's assumption aligns with these findings. The gas price data were sourced from the United States Energy Information Administration (EIA). The "hedonometer" [12] was used to gauge people's moods or happiness using Twitter data.

Another set of features relates to the COVID-19 pandemic, significantly impacting the restaurant industry. During this period, restaurants faced operational restrictions and a decline in dine-out willingness [17,62]. We collected epidemiological variables (e.g., confirmed cases and deaths) and policy measures (e.g., gathering restrictions and COVID testing policy) from the COVID-19 Data Hub [16]. In addition, we introduced a variable called "winter surges," representing the winter period (November 15 through January 31) characterized by the emergence of new variants and cold weather. Table 3 presents two types of feature sets (i.e., operational, time-related, and macroeconomic and pandemic features) and the specific features within each set.

Table 3
Types of Input and Specific Input Variables.

Input Types	Specific Input Variables
Operational Data & Time-related Features (OT)	Sales, Promotion Month, Week of the Month, Week of the Year, Holidays
Macroeconomic and Health Policy Features (MH)	Gas Prices, Covid Cases, Covid Deaths, Transport Closing, Stay Home Restrictions, Internal Movement Restrictions, Information Campaigns, Testing Policy, Facial Coverings, Vaccination Policy, Elderly People Protection, Hedonometer, CPI, Winter Surges

3.2. Demand forecasting models

Restaurant forecasting is considered a type of predictive analytics, and the quality of predictive analytics depends on data availability and quality. Restaurants vary in their capabilities for storing and managing the data for demand forecasting. This "data capability" is the key building block of analytics for accurate forecasting [5]. "Management decisions informed by these data analytic methods are only as good as the data on which they are based" [20].

3.2.1. Data preparation

The data from the industry partner included two features – past sales and promotion information. Exploratory data analysis was conducted for two objectives. First, the data quality, including inconsistencies and redundancies, was evaluated using various data wrangling and visualization techniques. Second, the original daily and menu-level sales data were aggregated into different levels, such as weekly and national sales, for trend analysis and anomaly detection. Thus, the raw data was transformed into several structures for different types of analysis (e.g., weekly, menu-level, and national).

3.2.2. Baseline forecast

The exploratory data analysis shows how the restaurant has forecasted customer demand. As noted earlier, there are several promotional periods each year with a combination of new and existing menu plates. Forecasts are made for each menu plate at the beginning of the promotion, and the forecast amount remains considerably unchanged, as shown in Fig. 4. This forecast is considered the baseline in this study for comparison with the ML- and DL-based forecasting models introduced in the following subsection. Fig. 4 illustrates the baseline forecast for one menu plate, introduced in January 2019 and promoted between March 2019–December 2019, using the initial demand forecast for that period.

3.2.3. ML- and DL-based forecasting models

Several ML/DL-based forecast models and their results are compared

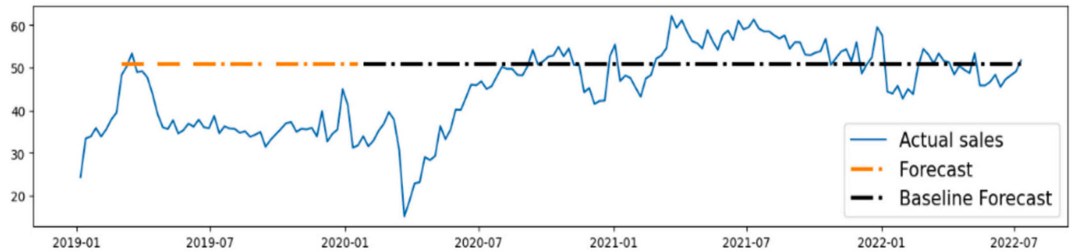

Fig. 4. Illustration of the Baseline Forecast.

with the baseline forecast. We considered two combinations of variables or features: (1) operational data and time-based features (OT features) and (2) operational data, time-based features, and macroeconomic/health-related features (OTMH features).

Six types of ML- and DL-based algorithms were considered – regularized regression, DT-based, SVR, MLP, RNN, and transformers. We developed models based on two regularized regression algorithms (Lasso and Ridge), which offer automatic feature selection and improved model explainability. Four DT algorithms (Random Forest, GBM, XGBoost, and Light GBM) were used to build the forecasting models. SVR was applied to the OT and OTMH feature combinations. Three types of DL-based methods, including NHITS (MLP), DeepAR (RNN), and TFT (transformers), were used to develop the demand forecasting models.

We implemented 20 forecasting models with two sets of data inputs (OT and OTMH) and 10 ML/DL algorithms (2×10). For instance, one of the simplest forecast models uses OT features with Lasso. Conversely, more complex models utilize OTMH. Fig. 5 depicts the ML/DL-based models considered in this research.

3.2.4. Training and evaluation

Forecasting customer demand is critical for effective supply chain planning in restaurant contexts. Multi-horizon forecasting is necessary to use the forecasts as input for supply chain planning. For training the models, we selected 28 menu plates whose lifespans lasted for at least 80 weeks. This criterion is based on the understanding that restaurant sales are sensitive to promotions and time-based factors, such as holidays. Therefore, training each ML and DL model with more than one year of operational data is necessary. Each trained model is evaluated by applying it to testing data, generating predicted demands for the 28 menu plates. For evaluation, the actual sales and the predicted demands were compared. The smaller the gap between the two values, the better the model performance.

We adopted the expanding window method for training and evaluating the models. The models were trained and evaluated using two expanding windows – the turbulent phase (January–March 2022) and the stable phase (April–June 2022). The extant literature [13,49] has long recognized two business environments – stable and turbulent. In a stable environment, external factors are relatively predictable, whereas a turbulent environment is highly volatile and uncertain. We can gain valuable insights into their robustness by assessing the models in both types of environments. This evaluation approach enhances the validity of the models' performance.

Fig. 6 describes the expanding window approach used in this study. In the first expanding window, when the restaurant chain's sales were turbulent due to a new COVID variant, the sales of this period were used to test the models' performance. To measure multi-horizon forecasting performance, each model generated forecast values for four to twelve weeks (4, 5, 6, 7, 8, 9, 10, 11, and 12 weeks). The models were evaluated using a second expanding window where the stable phase was used to test the models' quality. This ensures that the models are evaluated using contrasting circumstances.

This research assesses the value or effectiveness of external data and ML/DL models in restaurant demand management. Similar to previous studies, we access this value by comparing 20 demand forecasting models based on their accuracy. Several evaluation metrics exist for ML models, and model performance can vary depending on the metrics used

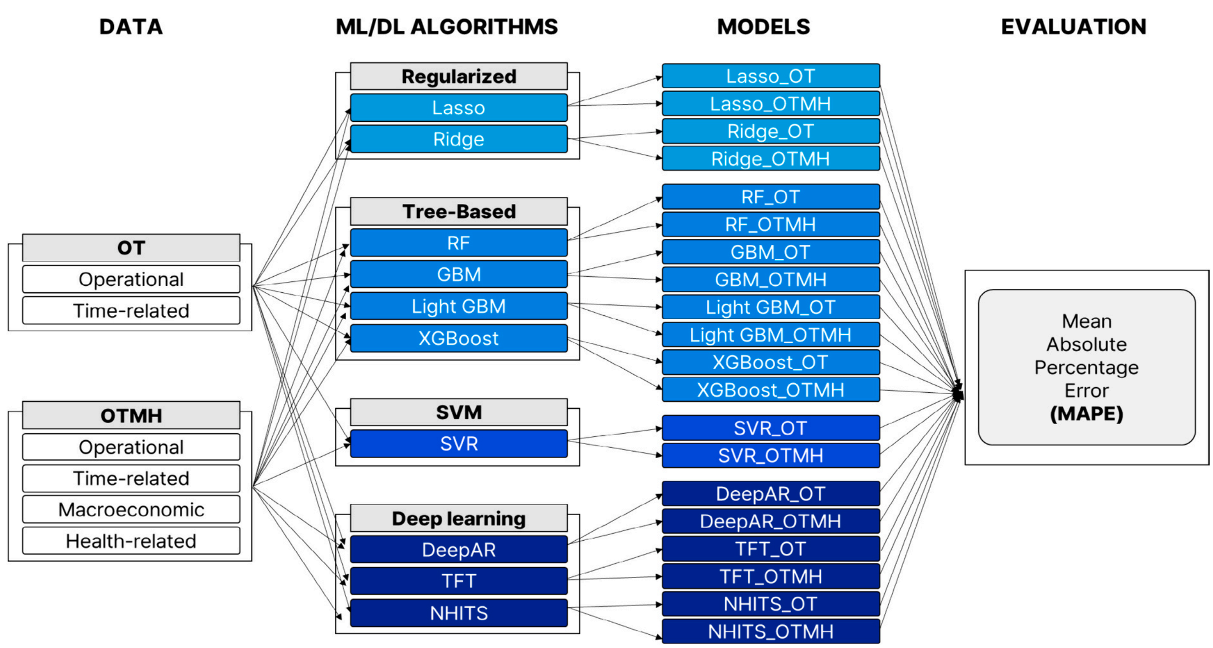

Fig. 5. The Framework of the ML- and DL-based Models.

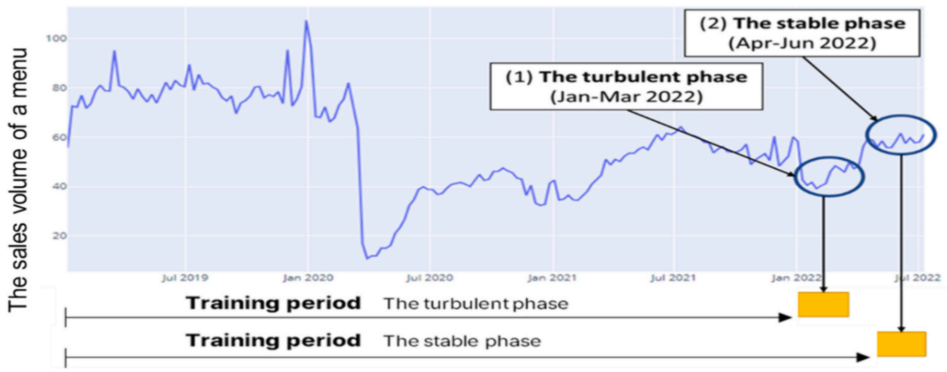

Fig. 6. Two Expanding Windows.

[3,7,59]. Among the most commonly used metrics is the Mean Absolute Percentage Error (MAPE), calculated as (1/n)xΣ(|actual–forecast|/|actual|)*100. MAPE is a scale-independent metric and offers intuitive interpretation, measuring the percentage difference between forecasted and actual values. Studies in demand forecasting often report forecasting model accuracy using MAPE [e.g., [11,38]]. We evaluated the baseline and 20 ML and DL models based on MAPE. For instance, we assess the value of external data (MH features) in two models (Model A with OT features and Model B with OTMH features). Assuming that the two models' MAPE is 0.38 and 0.32, respectively, we calculated that the value of external data is a 16% improvement in accuracy, obtained by dividing (0.38–0.32) by 0.38.

4. Results

This section reports the performance of the company's baseline and 20 ML- and DL-based models. The ML- and DL-based models were built based on two input variables – OT and OTMH features. The models were evaluated during two contrasting sales periods – turbulent (January–March 2022) and stable (April–June 2022). In each period, the models generated multi-horizon forecast values for four to twelve weeks (4, 5, 6, 7, 8, 9, 10, 11, and 12), and their performances were measured in terms of MAPE. We raised three research questions in the introduction, and the results are presented accordingly.

4.1. Comparison of OT with OTMH

Table 4 summarizes the advantages of the models with OTMH over those with OT features.

The relative advantage of OTMH over OT features is consistently significant in the ML models, irrespective of the algorithm types and the evaluation period. In the turbulent period, the relative improvement of the ML models ranges from 19% (Lasso) to 62% (XGBoost). The improvement is more pronounced in the stable period, ranging from 30% (Lasso) to 82% (XGBoost). For instance, the XGBoost model's average MAPE over multi-horizon forecasting (4–12 weeks) is 0.61 with OT features; however, it improves to 0.23 with OTMH features. This represents a 62% improvement in forecasting accuracy, attributable to the inclusion of MH features. Among the ML models, DT models show the most significant improvement compared to regularized regression and SVR. Fig. 7 illustrates the relative improvement of the DT models with OTMH features in both the turbulent (54%) and stable (73%) periods.

with the Highlight of the DT Models.

In Fig. 8, the performance of DL models with OTMH features displays a more complex relationship. During the turbulent period, incorporating external features, such as MH, did not enhance the performance of the DL models. The DL models with only OT features outperformed those with OTMH features and most other ML models in multi-horizon forecasting. However, the scenario shifted in the stable period. Adding MH features notably improved the DL models' accuracy by 17% for NHITS, 22% for TFT, and 45% for DeepAR.

4.2. Comparison of the ML and DL models with the baseline forecast

We estimated the company's baseline forecast using the sales promotion data obtained, including initial forecasts for all promotional menu items. Based on our estimations, the average MAPE of the company's baseline forecast for 28 menu items was approximately 1.60 and 2.43 in the turbulent and stable phases, respectively. Interpreting MAPE means, on average, the company's forecast deviated by over 160% and 240% from the actual sales in the turbulent and stable phases, respectively.

Table 5 summarizes the results: (1) the accuracy of the ML/DL models in terms of MAPE and (2) the models' relative improvement over the baseline forecast. Regardless of the input features, all the models significantly outperformed the company's estimated baseline forecast in turbulent and stable evaluation phases. However, the extent of MAPE improvement over the company's baseline forecast varied depending on the evaluation period, the type of input data, and the algorithms used. During the turbulent period, the most significant improvement (90%) was observed with NHITS with OT features. NHITS with OTMH features offered a 95% improvement, the highest during the stable period.

MAPE*: The average MAPE of multi-horizon forecasting (4–12 weeks).

RI**: Relative improvement of the ML/DL model over the baseline forecast.

Fig. 9 shows that during the stable phase (April–June 2022), the accuracy of the ML/DL models in terms of MAPE ranged from 0.80 (XGBoost) to 0.12 (NHITS), compared to approximately 2.43 for the baseline. All the ML/DL models based on OTMH provided more improvement than those with OT features. With OT features, the accuracy ranged from 0.80 (XGBoost) to 0.15 (NHITS), leading to an average relative MAPE improvement over the company's baseline forecast of 67–93%. With OTMH features, the accuracy ranged from 0.49 (SVR) to 0.12 (NHITS), resulting in an average improvement of 79–94%.

4.3. Comparison of the DL and ML models

The DL models significantly outperformed the ML models in turbulent and stable periods with OT features. Fig. 10 compares the DL and the ML models using OT features. In the turbulent period, the accuracy of the DL models for multi-horizon forecasting was 0.16 for NHITS, 0.17 for TFT, and 0.31 for DeepAR. In contrast, the accuracy of the ML models ranged from 0.61 (XGBoost) to 0.41 (Lasso and Ridge). For instance, NHITS demonstrated a 73% improvement in MAPE over XGBoost. During the stable period, the accuracy of the DL models was 0.16 (NHITS), 0.20 (TFT), and 0.27 (DeepAR), while the ML models showed

Table 4
The Advantage of OTMH over OT.

				4	5	6	7	8	9	10	11	12
The Turbulent Phase	Regularized regression	Lasso	OT	0.38	0.37	0.38	0.4	0.41	0.43	0.44	0.46	0.47
			OTMH	0.32	0.24	0.26	0.29	0.31	0.32	0.32	0.37	0.65
			Improvements	**16%**	**35%**	**32%**	**28%**	**24%**	**26%**	**27%**	**20%**	**-38%**
		Ridge	OT	0.38	0.37	0.38	0.39	0.41	0.43	0.44	0.46	0.47
			OTMH	0.25	0.24	0.25	0.27	0.3	0.31	0.3	0.33	0.57
			Improvements	**34%**	**35%**	**34%**	**31%**	**27%**	**28%**	**32%**	**28%**	**-21%**
	Decision trees	RF	OT	0.53	0.53	0.54	0.56	0.56	0.57	0.58	0.58	0.58
			OTMH	0.17	0.18	0.18	0.23	0.25	0.26	0.28	0.37	0.54
			Improvements	**68%**	**66%**	**67%**	**59%**	**55%**	**54%**	**52%**	**36%**	**7%**
		GBM	OT	0.48	0.49	0.5	0.53	0.54	0.57	0.57	0.59	0.59
			OTMH	0.22	0.21	0.19	0.21	0.24	0.23	0.23	0.25	0.52
			Improvements	**54%**	**57%**	**62%**	**60%**	**56%**	**60%**	**60%**	**58%**	**12%**
		LightGBM	OT	0.46	0.47	0.48	0.5	0.48	0.5	0.52	0.55	0.57
			OTMH	0.14	0.16	0.15	0.16	0.16	0.17	0.22	0.33	0.5
			Improvements	**70%**	**66%**	**69%**	**68%**	**67%**	**66%**	**58%**	**40%**	**12%**
		XGBoost	OT	0.58	0.58	0.58	0.59	0.62	0.64	0.64	0.66	0.65
			OTMH	0.18	0.18	0.13	0.19	0.2	0.25	0.25	0.27	0.47
			Improvements	**69%**	**69%**	**78%**	**68%**	**68%**	**61%**	**61%**	**59%**	**28%**
	SVM	SVR	OT	0.49	0.49	0.5	0.52	0.51	0.53	0.53	0.54	0.55
			OTMH	0.29	0.29	0.3	0.31	0.32	0.34	0.35	0.36	0.37
			Improvements	**41%**	**41%**	**40%**	**40%**	**37%**	**36%**	**34%**	**33%**	**33%**
	Deep learning	DeepAR	OT	0.29	0.3	0.28	0.3	0.29	0.31	0.3	0.34	0.42
			OTMH	0.32	0.24	0.19	0.2	0.22	0.23	0.23	0.24	0.28
			Improvements	**-10%**	**20%**	**32%**	**33%**	**24%**	**26%**	**23%**	**29%**	**33%**
		TFT	OT	0.09	0.13	0.15	0.18	0.17	0.14	0.14	0.26	0.3
			OTMH	0.17	0.2	0.27	0.17	0.25	0.18	0.3	0.19	0.45
			Improvements	**-89%**	**-54%**	**-80%**	**6%**	**-47%**	**-29%**	**-114%**	**27%**	**-50%**
		NHITS	OT	0.14	0.1	0.19	0.12	0.17	0.14	0.15	0.21	0.28
			OTMH	0.24	0.16	0.21	0.18	0.35	0.25	0.62	0.34	0.41
			Improvements	**-71%**	**-60%**	**-11%**	**-50%**	**-106%**	**-79%**	**-313%**	**-62%**	**-46%**
The Stable Phase	Regularized regression	Lasso	OT	0.51	0.51	0.50	0.50	0.51	0.51	0.52	0.52	0.52
			OTMH	0.32	0.32	0.33	0.33	0.35	0.37	0.38	0.40	0.41
			Improvements	**38%**	**36%**	**35%**	**34%**	**31%**	**27%**	**27%**	**24%**	**21%**
		Ridge	OT	0.51	0.51	0.51	0.50	0.51	0.51	0.51	0.52	0.52
			OTMH	0.31	0.30	0.31	0.31	0.33	0.36	0.36	0.38	0.40
			Improvements	**41%**	**41%**	**38%**	**38%**	**35%**	**31%**	**29%**	**26%**	**23%**
	Decision trees	RF	OT	0.72	0.69	0.68	0.68	0.68	0.69	0.69	0.68	0.67
			OTMH	0.12	0.12	0.12	0.12	0.12	0.14	0.15	0.17	0.18
			Improvements	**83%**	**83%**	**83%**	**82%**	**83%**	**79%**	**79%**	**75%**	**73%**
		GBM	OT	0.61	0.62	0.61	0.61	0.61	0.61	0.62	0.61	0.61
			OTMH	0.15	0.15	0.17	0.17	0.15	0.20	0.19	0.23	0.27
			Improvements	**76%**	**76%**	**73%**	**72%**	**75%**	**67%**	**69%**	**62%**	**56%**
		LightGBM	OT	0.57	0.58	0.57	0.58	0.58	0.58	0.59	0.59	0.58
			OTMH	0.17	0.20	0.21	0.20	0.18	0.21	0.24	0.26	0.31
			Improvements	**69%**	**66%**	**64%**	**66%**	**68%**	**63%**	**59%**	**56%**	**47%**
		XGBoost	OT	0.84	0.80	0.79	0.79	0.80	0.82	0.82	0.80	0.78
			OTMH	0.12	0.12	0.11	0.14	0.12	0.15	0.16	0.19	0.22
			Improvements	**86%**	**85%**	**86%**	**83%**	**85%**	**82%**	**80%**	**76%**	**72%**
	SVM	SVR	OT	0.70	0.71	0.69	0.68	0.68	0.68	0.68	0.67	0.65
			OTMH	0.48	0.48	0.48	0.48	0.49	0.50	0.51	0.51	0.51
			Improvements	**32%**	**32%**	**31%**	**29%**	**28%**	**27%**	**26%**	**23%**	**22%**
	Deep learning	DeepAR	OT	0.24	0.24	0.26	0.25	0.24	0.29	0.28	0.30	0.32
			OTMH	0.10	0.11	0.13	0.15	0.15	0.20	0.16	0.16	0.17
			Improvements	**58%**	**55%**	**48%**	**39%**	**35%**	**33%**	**44%**	**45%**	**48%**
		TFT	OT	0.12	0.21	0.21	0.24	0.20	0.24	0.21	0.19	0.21
			OTMH	0.11	0.12	0.14	0.18	0.13	0.15	0.24	0.17	0.17
			Improvements	**9%**	**42%**	**33%**	**24%**	**34%**	**37%**	**-12%**	**11%**	**20%**
		NHITS	OT	0.18	0.11	0.14	0.15	0.13	0.17	0.13	0.21	0.19
			OTMH	0.14	0.11	0.10	0.14	0.12	0.16	0.09	0.13	0.15
			Improvements	**23%**	**4%**	**24%**	**7%**	**9%**	**5%**	**30%**	**36%**	**16%**

higher MAPE values, including 0.80 (XGBoost), 0.69 (RF), 0.68 (SVR), and 0.51 (Lasso and Ridge). Consequently, NHITS represented an 80% improvement in accuracy over XGBoost.

Fig. 11 compares the DL and ML models using OTMH features. In this turbulent period, the DL models had no distinct advantage over the ML models with OTMH features. However, DL models exhibit slightly better performance for long-term forecasting (e.g., 11th and 12th weeks) than their ML counterparts. During the stable period, the advantage MAPE for the three DL models becomes more apparent in multi-horizon forecasting. The average MAPE of the three DL models is 0.16, contrasting with the 0.24 average MAPE for the four DT models. This indicates a pronounced superiority of the DL models in stable conditions, particularly for longer forecasting horizons.

5. Discussions and implications

The main objectives of this study were to answer three research questions: (1) the value of incorporating external data (MH) alongside OT features over OT features alone, (2) the benefits of using advanced analytics, specifically ML and DL, in the context of demand forecasting,

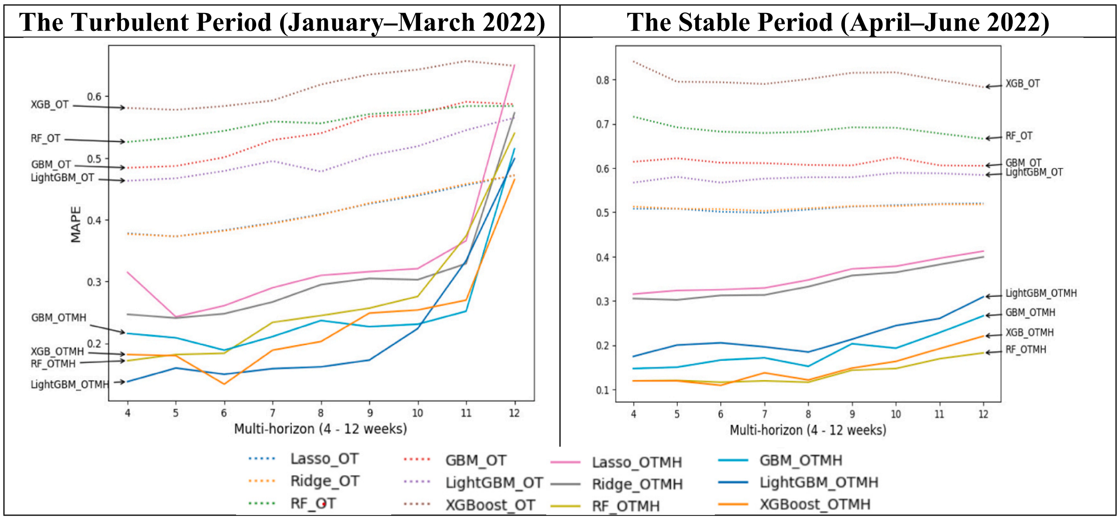

Fig. 7. The Relative Improvement of the ML Models with OTMH Features.

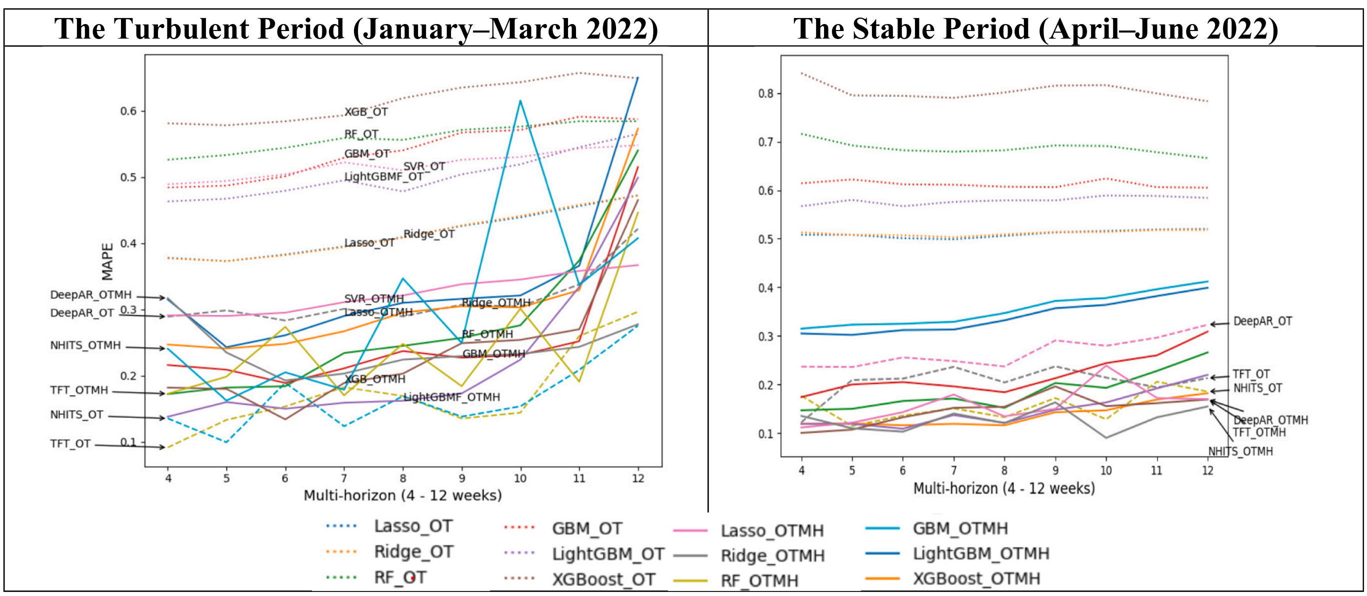

Fig. 8. The Relative Improvement of the DL Models with OTMH Features.

Table 5
The Advantage of ML/DL Models Over the Baseline Forecast.

		The Turbulent Period				The Stable Period			
		OT		OTMH		OT		OTMH	
		MAPE*	RI**	MAPE*	RI**	MAPE*	RI**	MAPE*	RI**
Regularized Regression	Lasso	0.41	**74%**	0.34	**78%**	0.51	**79%**	0.35	**85%**
	Ridge	0.41	**74%**	0.31	**80%**	0.51	**79%**	0.34	**86%**
Decision Tree	RF	0.55	**65%**	0.27	**83%**	0.68	**72%**	0.13	**94%**
	GBM	0.53	**67%**	0.25	**84%**	0.61	**75%**	0.18	**92%**
	LightGBM	0.50	**50%**	0.22	**86%**	0.57	**76%**	0.22	**91%**
	XGBoost	0.61	**62%**	0.23	**85%**	0.80	**67%**	0.14	**94%**
SVM	SVR	0.51	**68%**	0.23	**86%**	0.68	**72%**	0.49	**80%**
Deep Learning	DeepAR	0.31	**80%**	0.32	**80%**	0.26	**89%**	0.14	**94%**
	TFT	0.17	**89%**	0.24	**85%**	0.20	**92%**	0.15	**94%**
	NHITS	0.16	**90%**	0.30	**81%**	0.15	**94%**	0.12	**95%**

and (3) the value of DL models over ML models.

5.1. The value of OTMH over OT

The effectiveness of demand forecasting models is contingent on the data quality [20]. This is particularly true for models based on ML and

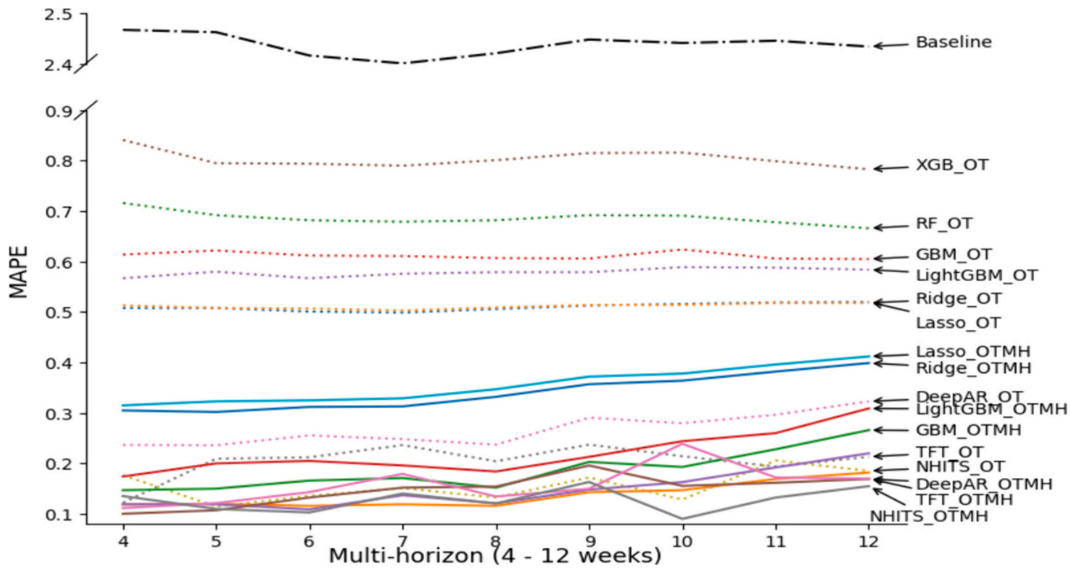

Fig. 9. ML/DL Models' Accuracy (MAPE) in the Stable Phase Compared to the Baseline Forecast.
Note: Models utilizing OT features are represented with dotted lines, while those with OTMH features are shown with solid lines.

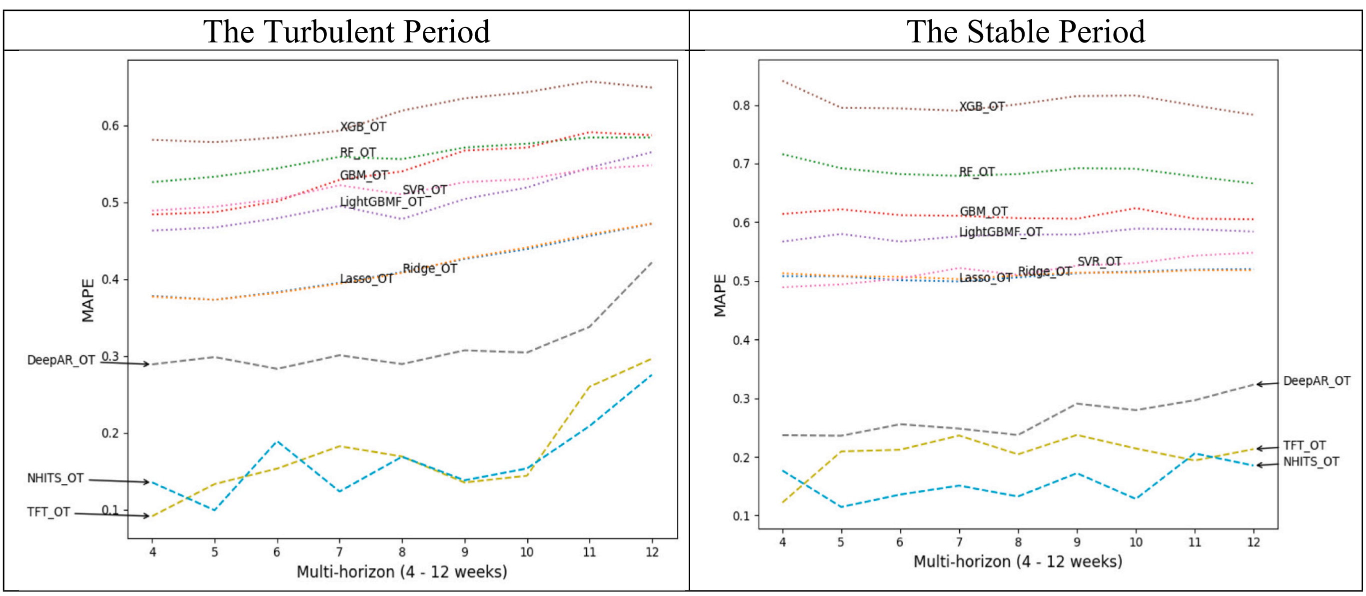

Fig. 10. Comparison of DL and ML Models Using OT Features in Turbulent and Stable Periods.

DL algorithms, which benefit from including more relevant features. Our study compared models using limited OT features with additional MH features. While OT features are commonly used in demand forecasting, our findings suggest that additional external features are necessary in dynamic environments, such as during a pandemic.

The study demonstrates that the value of including MH features varies, and it is necessary to consider both the algorithms and the degree of turbulence during the prediction period. Fig. 12 summarizes our discussion on the value of adding MH features. The most substantial improvement was observed with the ML models, particularly in stable environments and, to a moderate extent, turbulent environments. This suggests that restaurants should consider incorporating external variables, such as MH, when adopting ML models for demand forecasting. In contrast, the improvement observed in the DL models after adding MH features is negligible during the turbulent period and only moderately better in the stable environment. This suggests that the inclusion of MH features does not significantly enhance the performance of DL models in rapidly changing conditions. However, it is essential to note that DL models relying on OT features have demonstrated a strong performance, particularly in the turbulent period. This indicates the robustness of the DL models in handling dynamic market conditions even with a limited set of features.

There are two practical implications. First, restaurants must weigh the costs and benefits when choosing features and models for demand forecasting. OT features are readily accessible; however, acquiring external MH features demands analytical expertise and resources. These resources are often scarce in small and medium-sized restaurants. In such cases, using DL models is particularly advantageous as they can offer competitive forecasting accuracy with only OT features. ML models, especially DT-based models, are viable for large businesses with the necessary resources. These models significantly improve accuracy with the addition of MH features. Given the increasing complexity and diversity of risks in the restaurant industry [60], identifying external variables related to MH is crucial, especially for large chains [27].

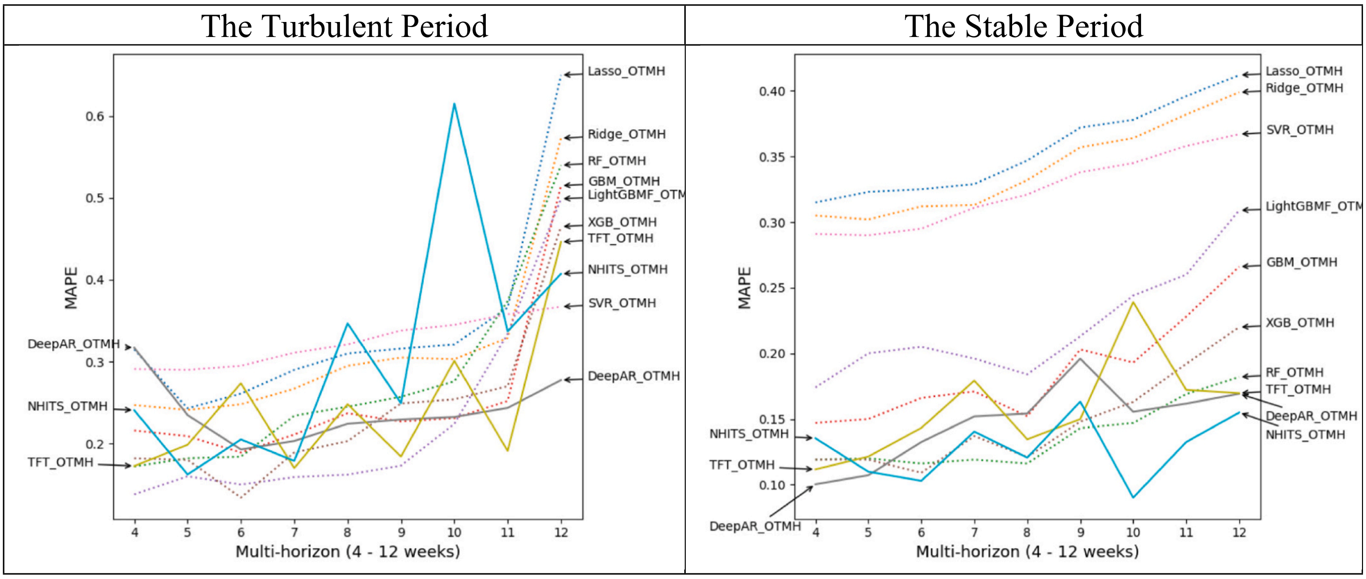

Fig. 11. Comparison of the DL and ML models with OTMH features.

ALGORITHM		
DL	Moderate performance	Low performance
ML	High performance	Moderate performance
	Stable	Turbulent

ENVIRONMENT (Testing period)

Fig. 12. The Value of Adding MH Features.

Beyond COVID-19, restaurants face various challenges, such as food safety, brand sabotage, dysfunctional customer behavior, and employee service sabotage, which directly impact customer demand [22,29,31,40]. Social media has become a valuable source of data for environmental scanning. Data from social media (e.g., reviews, photos, and sentiments) can enhance demand forecasting accuracy [11,28,63].

In addition, restaurants should focus on high-quality internal data, including attributes of menu items and shipment data for similar items. Practices, such as product master data management (PMDM) in consumer goods and manufacturing [61], can be adopted to improve data quality for effective demand forecasting. Such internal data are particularly valuable for forecasting the demands of new menu items using data from existing items [30].

5.2. *The value of ML/DL models over the baseline forecast*

The restaurant industry has been slow to adopt advanced data and analytics for decision support and operations management [52]. Most restaurants currently utilize tools such as Excel spreadsheets, and there is less satisfaction with the forecasting process and quality among those solely relying on such standalone tools [2]. One of the significant challenges of using conventional tools (e.g., spreadsheets) for demand forecasting is the lack of a streamlined process between data generation and preparation, as well as model development and deployment. In short, it is difficult to achieve a high level of automation in demand forecasting, resulting in infrequent forecasting updates and ineffective forecasting practices. This explains the current demand forecasting practices within the restaurant chain introduced in this study.

The results indicate that using ML and DL models, based on either OT or OTMH features, can significantly improve demand forecasting. Regardless of analytical techniques or algorithms used, systematically including internal (e.g., marketing/promotion) and external (e.g., macroeconomic indicators) data in DL/ML model development can enhance demand forecasting practices and outperform the company's estimated baseline forecast in turbulent and stable phases. However, the relative improvement varies depending on three factors – forecasting periods, types of input features, and algorithms used.

Overall, the most accurate model is with the NHITS, a DL algorithm, using OT features in the turbulent period and OTMH features in the stable period. The most significant improvement, between 85 and 95%, is observed with OTML features when forecasting during the stable period, regardless of the ML/DL algorithm. The least improvement, ranging from 50 to 90%, occurs when the models are built with OT features for the turbulent period. ML models benefit from adding MH features along with OT features, and this positive effect is more significant in the stable period for DT algorithms than regularized algorithms. Such differences may be due to the assumptions shared by the two families of models [19]. Regularized regression assumes a linear relationship between input features and sales; whereby the potential model overfitting is handled by $\ell1$ (Lasso) and $\ell2$ (Ridge) norms. "Regularization" aims for complexity reduction in demand forecasting models. DT-based models assume non-linearity in the relationship between input features and sales. Thus, when input variables have non-linear relationships with sales, the results of regularized regression models are suboptimal, while DT-based models are likely to outperform. Similarly, DL models outperformed ML models, including SVR and DT, as they are adept at finding complex, non-linear relationships between input features and target variables.

The ML and DL models are more complex than conventional demand forecasting models. Recent literature recognizes the importance of explainability in such models. Regularization and DT-based ML models are inherently explainable [42]. Regularized regression models provide feature (or variable) selection by penalizing the size of estimated coefficients (typically zero) of unimportant features, and DT-based models generate information about which features (or variables) are essential based on different measures, such as "weight" and "gain" [39].

5.3. *The value of the state-of-art DL models over ML models*

While the extant literature offers limited discussion on using the state-of-the-art DL models in restaurant demand forecasting contexts, DL is increasingly applied for time series forecasting. This often involves global learning, enabling the models to learn patterns across time series

[21]. DL algorithms, such as NHITS, TFT, and DeepAR used in this study, are developed explicitly for time series problems and are adept at capturing various patterns, including trends and seasonality. These characteristics give them an edge over traditional ML models in terms of performance.

This study demonstrates that DL models either closely match or outperform traditional ML models in restaurant demand forecasting. Their advantage over ML models varies significantly depending on input variables (OT vs OTMH) and forecasting periods (stable vs turbulent). The findings are summarized in Fig. 13. When only OT variables are available, DL models significantly outperform ML models in stable and turbulent environments. With the addition of MH features, while ML models' performance improves substantially, DL models still moderately outperform or match ML models in both environments.

The results indicate that DL models are less sensitive to the number of input variables than ML models. They demonstrate strong or comparable performance, regardless of additional variables. For instance, with only OT features, the NHITS model – one of the best-performing DL models – shows over a 70–80% improvement over DT-based ML models, such as XGBoost. This is largely attributed to the unique architectures of the DL models, which are tailored for time series data and are adept at capturing patterns, such as seasonal trends. In addition, individual DL algorithms employ specific mechanisms, such as backcasting in NHITS [6]. With the inclusion of MH variables, the performance advantage of DL models over ML models decreases to 33% in stable periods and becomes marginal in turbulent periods. This suggests that ML models require a more significant number of robust predictors, and with such input features, their performance can rival DL models. This finding is essential for restaurants, particularly those with limited access to various internal and external variables for demand forecasting. DL models emerge as a valuable analytical tool for these restaurants, irrespective of the business environment.

While this study highlights the competitive advantage of DL models in terms of accuracy, it notes that advanced DL models have complex architectures and, thus, are lower in intrinsic interpretability than ML models. Efforts to improve interpretability include methods such as SHAP (Shapley Additive exPlanations) [32] and LIME (Local Interpretable Model-agnostic Explanations) [44]. Restaurants, therefore, might face a trade-off between performance and interpretability in demand forecasting.

6. Conclusion and future directions

Demand forecasting in the restaurant industry lags behind other manufacturing and consumer goods sectors. Research on restaurant supply chains is scant in decision sciences, operations management, and hospitality literature [45,52]. The restaurant industry faces increasingly diverse challenges, and large chains have adopted digitalization to enhance supply chain visibility and response to external events. However, the pandemic presented an unprecedented challenge to restaurants. In this context, the utilization of advanced analytics, including the expansion of data capability and the application of ML and DL, gained heightened attention. In collaboration with a leading United States restaurant chain, this study explored the potential of enhanced data

INPUT TYPE	Stable	Turbulent
OTMH	Moderately outperforming to closely matching	Closely matching
OT	Significantly outperforming	Significantly outperforming

ENVIRONMENT (Testing period)

Fig. 13. The Performance of the State-of-Art DL Models over Traditional ML Models.

access and ML/DL algorithms for demand forecasting in stable and turbulent scenarios. To our knowledge, this is the first empirical study combining two data sources, several ML and state-of-the-art DL algorithms, and two testing scenarios. Our findings contribute significantly to the academic literature and industry practices in restaurant analytics and demand forecasting.

This study explored the demand forecasting models within the restaurant industry, driven by three RQs based on the Dynamic Capabilities Framework. RQ1 delved into enhancing sensing capability through various internal and external factors. This study examined multiple input features, including OTMH features and sales and promotional data from January 2019–July 2022. We found a significant enhancement in forecasting accuracy by integrating external macroeconomic and pandemic-related variables, particularly with the ML models. This improvement is notable in models utilizing OTMH features consistently outperforming those with only OT features across turbulent and stable periods. This implies that incorporating external data can enhance the sensing capability so that restaurants can anticipate the changes in demand more accurately and adjust their strategies proactively, which aligns with the critical aspect of the Dynamic Capabilities Framework.

To answer RQ2 and RQ3, which suggest the usefulness of advanced analytics in seizing and transforming capabilities, the comparative analysis of ML and DL models was performed. While ML models showed notable improvement with the inclusion of OTMH features, DL models excelled in handling dynamic market conditions even with limited data inputs. This difference revealed the potential of advanced analytics in seizing accurate forecasting opportunities and adjusting operational practices to align with anticipated market trends.

The findings from this study can provide valuable insights for future research on decision support. As more data becomes available, future research on decision support systems should take into account a wide range of input features, including menu characteristics, weather, consumer indices, competitor information, and social media [e.g., 11, 28, 63]. Since the restaurant environment is complex and carries diverse risks [60], it is important to consider external variables related to macroeconomic environments and risks to propose effective decision-making strategies [27]. Additionally, future research on decision support systems should encompass a broader span of sales and promotional data. ML/DL models are expected to perform better with more input features and larger sample sizes [4].

The practical implications for decision-makers are also significant. ML/DL for time series forecasting is rapidly evolving, with emerging algorithms, particularly in the DL domain, for demand forecasting [e.g., [58]]. Our study suggests that DL models, even with a limited number of OT features, offer the opportunity to build highly accurate forecasting models due to their intrinsic mechanisms, such as autoregression. As the restaurant industry lags in data capability compared to consumer goods and manufacturing sectors, decision-makers in restaurant sectors should implement these cutting-edge DL-based time series techniques and methods to improve operational efficiency.

While this study offers valuable insights, it has several limitations. This study utilized data from a single large restaurant chain. The findings may not be generalizable because of the variability in internal features, such as main ingredients, menus, and customer demographics. Furthermore, it contextualized a unique global health risk within the COVID-19 pandemic period. The specific impacts observed during this time may not be directly applicable to other turbulent periods that may influence future macroeconomic conditions.

Declaration of generative AI in scientific writing

During the preparation of this work the author(s) used ChatGPT 3.5 in order to improve readability and language. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

Funding source

N/A.

Data visualization

Color should be used for all figures in manuscript.

CRediT authorship contribution statement

Bongsug (Kevin) Chae: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Chwen Sheu:** Writing – review & editing, Data curation, Conceptualization. **Eunhye Olivia Park:** Writing – review & editing, Visualization, Conceptualization.

Declaration of competing interest

While we obtained data from a US restaurant chain, we did not receive any financial benefits from the company, nor did we advocate for their interests. Therefore, we have no conflicts of interest to disclose.

Data availability

Data not available due to commercial restrictions.

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